# D0. Choosing an analysis: start here

Answer four questions in order. The blue labels point to the focused diagrams D1 to D6.

**1. What is the question?**
Write it in one sentence first

- **Describe**
  What do the data look like?
  D1
- **Compare groups**
  Do groups differ in the outcome?
  D2 continuous, ordinal
  D3 categorical
- **Assess an association**
  Is X related to Y, possibly adjusted for others?
  D3 both categorical
  D4 otherwise
- **Predict**
  Estimate Y for new people
  D4 plus a train/test split
- **Discover groups**
  Are there subgroups? No outcome variable
  D5

**2. What type is the outcome?**
The variable you measure or explain

- **Continuous**
  Measured on a scale
  e.g. SBP, BMI, CRP
- **Binary**
  Two categories
  e.g. high BP yes/no
- **Nominal**
  Unordered categories
  e.g. district
- **Ordinal**
  Ordered categories
  e.g. satisfaction 1 to 5
- **Count**
  Number of events
  e.g. clinic visits
- **Time-to-event**
  Time until an event; some not yet observed (censored)

**3. How were the observations collected?**

- **Independent**
  One measurement per person; people unrelated
  D2 to D4
- **Paired**
  Same people twice, or matched pairs
  paired methods in D2, D3
- **Repeated**
  Three or more measurements per person
  D6 before analysing
- **Clustered**
  People grouped in clinics, households, schools
  D6 before analysing

**4. How many groups or predictors?**

- **Two groups**
  e.g. women and men; t-test, Wilcoxon, 2 x 2 table
- **Three or more groups**
  e.g. four districts; ANOVA, Kruskal-Wallis, r x c table
- **One predictor**
  Correlation or simple regression
- **Several predictors**
  Adjusted associations: multiple, logistic or Poisson regression

**Use these diagrams to structure your thinking, not as automatic rules.** Look at the data with plots, think about how they were collected, and report an effect estimate (difference, ratio, slope) with its 95% confidence interval, not only a p-value. Sample size alone does not decide between a t-test and a rank-based test, and neither does the p-value of a normality test.